

Practical Tips for Solving Machine Learning Datasets

These tips apply to both the **Classification datasets** (Titanic, Heart Disease, IMDb, AG News, etc.) and **Regression datasets** (House Prices, California Housing, Insurance Cost, etc.).

1. Understand the Problem First

Before writing any code:

For Classification

Ask:

- What is the target column?
- Binary or multi-class?
- Balanced or imbalanced classes?

Examples:

- Titanic → Survived/Not Survived
- IMDb → Positive/Negative
- Iris → 3 flower species

For Regression

Ask:

- What numerical value are we predicting?

Examples:

- House Prices → Sale price
 - Insurance → Medical cost
 - Airbnb → Rental price
-

2. Perform Exploratory Data Analysis (EDA)

Always start with EDA.

Check

```
1 df.shape
2 df.info()
3 df.describe()
4 df.head()
```

Visualizations

Classification

- Count plots
- Class distribution
- Correlation heatmap
- Feature importance

Regression

- Histogram of target

- Scatter plots
- Correlation matrix
- Box plots

Useful libraries:

```
1 matplotlib
2 seaborn
3 plotly
```

3. Handle Missing Values

Numerical Features

```
1 Median
2 Mean
3 KNN Imputation
```

Categorical Features

```
1 Mode
2 "Unknown" category
```

Avoid simply dropping rows unless missing values are very few.

4. Handle Outliers

Important for regression datasets.

Common methods:

IQR

1	Q1
2	Q3
3	IQR

Z-Score

1	$ z > 3$
---	-----------

Robust Scaling

Useful when outliers exist.

Datasets where outliers matter:

- House Prices
 - Insurance Charges
 - Airbnb Prices
 - Diamond Prices
-

5. Feature Engineering

Many competitions are won through feature engineering.

Example

Age:

- 1 Age
- 2 Age_Group
- 3 Age²

Date:

- 1 Year
- 2 Month
- 3 Day
- 4 Weekday
- 5 Quarter

Text:

- 1 Word Count
- 2 Character Count
- 3 TF-IDF Features

Images:

- 1 Resize
 - 2 Normalize
 - 3 Data Augmentation
-

6. Encode Categorical Variables

Label Encoding

Good for:

```
1 Low
2 Medium
3 High
```

One-Hot Encoding

Good for:

```
1 Gender
2 City
3 Country
4 Department
```

Tools:

```
1 sklearn.preprocessing
2 pd.get_dummies()
```

7. Scale Numerical Features

Required for:

- Logistic Regression
- SVM

- KNN
- Neural Networks

Use:

```
1 StandardScaler
2 MinMaxScaler
3 RobustScaler
```

Not necessary for:

- Random Forest
 - XGBoost
 - LightGBM
 - CatBoost
-

8. Split the Dataset Correctly

Typical split:

```
1 70% Train
2 15% Validation
3 15% Test
```

or

```
1 80% Train
2 20% Test
```

Use:

```
1 train_test_split()
```

For small datasets:

```
1 K-Fold Cross Validation
```

9. Start with Baseline Models

Never start with deep learning.

Try:

Classification

```
1 Logistic Regression
2 Decision Tree
3 Random Forest
```

Regression

```
1 Linear Regression
2 Decision Tree Regressor
3 Random Forest Regressor
```

10. Compare Multiple Algorithms

Classification

Test:

- 1 Logistic Regression
- 2 Naive Bayes
- 3 Random Forest
- 4 XGBoost
- 5 LightGBM
- 6 CatBoost
- 7 SVM

Regression

Test:

- 1 Linear Regression
- 2 Ridge
- 3 Lasso
- 4 ElasticNet
- 5 Random Forest
- 6 XGBoost
- 7 LightGBM
- 8 CatBoost

Create a comparison table.

11. Choose the Correct Evaluation Metric

Classification Metrics

Binary

- 1 Accuracy
- 2 Precision
- 3 Recall
- 4 F1 Score
- 5 ROC-AUC

Imbalanced Dataset

Focus on:

- 1 Precision
- 2 Recall
- 3 F1
- 4 PR-AUC

Examples:

- Fraud Detection
 - Churn
 - Medical Diagnosis
-

Regression Metrics

Common

- 1 MAE
- 2 MSE
- 3 RMSE
- 4 R^2

Recommended

- 1 MAE
 - 2 RMSE
 - 3 R^2
-

12. Handle Class Imbalance

Common datasets:

- Churn
- Fraud Detection
- Credit Approval

Techniques:

Oversampling

- 1 SMOTE
- 2 ADASYN

Undersampling

```
1 Random Under Sampling
```

Class Weights

```
1 class_weight="balanced"
```

13. Use Cross Validation

Instead of one train-test split:

```
1 5-Fold CV  
2 10-Fold CV
```

Benefits:

- More reliable evaluation
 - Less variance
 - Better model selection
-

14. Hyperparameter Tuning

Use:

```
1 GridSearchCV  
2 RandomizedSearchCV
```

```
3  Optuna
```

Tune:

Random Forest

```
1  n_estimators
2  max_depth
3  min_samples_split
```

XGBoost

```
1  learning_rate
2  max_depth
3  subsample
```

15. Feature Selection

Remove weak features.

Methods:

```
1  Correlation
2  Mutual Information
3  Recursive Feature Elimination
4  SHAP
```

Benefits:

- Faster training
- Better interpretability

- Less overfitting
-

16. Use Ensemble Models

Very effective for Kaggle.

Examples:

- 1 Random Forest
- 2 XGBoost
- 3 LightGBM
- 4 CatBoost
- 5 Voting
- 6 Stacking
- 7 Blending

Often outperform single models.

17. Dataset-Specific Tips

Tabular Data

Examples:

- Titanic
- House Prices

- Churn
- Insurance

Best Models:

- 1 CatBoost
 - 2 LightGBM
 - 3 XGBoost
 - 4 Random Forest
-

Text Classification

Examples:

- IMDb
- AG News
- Emotion

Start with:

- 1 **TF-IDF** + Logistic Regression

Then try:

- 1 **BERT**
 - 2 RoBERTa
 - 3 DistilBERT
-

Image Classification

Examples:

- MNIST
- Fashion MNIST
- Dogs vs Cats

Start with:

```
1 CNN
```

Then:

```
1 ResNet
2 EfficientNet
3 Vision Transformer
```

18. Prevent Overfitting

Symptoms:

- Excellent training score
- Poor test score

Solutions:

```
1 Cross-validation
2 Regularization
3 Dropout
```


- | |
|---------------------|
| 4 Early stopping |
| 5 Feature selection |
| 6 More data |
-

19. Interpret the Model

Tree-based Models

Use:

- | |
|----------------------|
| 1 Feature Importance |
|----------------------|

Any Model

Use:

- | |
|------------------------|
| 1 SHAP |
| 2 LIME |

Understand:

- Why model predicts
 - Important features
 - Business insights
-

20. Build a Complete Machine Learning Pipeline

A professional workflow:

```
1 Data Collection
2     ↓
3 EDA
4     ↓
5 Data Cleaning
6     ↓
7 Feature Engineering
8     ↓
9 Train/Test Split
10    ↓
11 Baseline Model
12    ↓
13 Model Comparison
14    ↓
15 Hyperparameter Tuning
16    ↓
17 Final Model
18    ↓
19 Evaluation
20    ↓
21 Deployment
```

Recommended Model Order

Classification

- | | |
|---|------------------------|
| 1 | 1. Logistic Regression |
| 2 | 2. Random Forest |
| 3 | 3. XGBoost |
| 4 | 4. LightGBM |
| 5 | 5. CatBoost |
| 6 | 6. Neural Networks |

Regression

- | | |
|---|----------------------------|
| 1 | 1. Linear Regression |
| 2 | 2. Random Forest Regressor |
| 3 | 3. XGBoost Regressor |
| 4 | 4. LightGBM Regressor |
| 5 | 5. CatBoost Regressor |
| 6 | 6. Deep Neural Networks |

Golden Rule

For **tabular datasets**, try **CatBoost, XGBoost, and LightGBM** before deep learning. In many Kaggle competitions, these models outperform neural networks on structured data while training much faster.